

SOP: Handling ServiceNow Incidents for Airflow Issues (AC360 / BDP 501 Data Processing)

Version: 1.0

Owner: BDP (Airflow & Data Processing Infrastructure)

1) Purpose

To provide a standardized procedure for triaging and resolving Airflow-related incidents impacting AC360 / BDP 501 Data Processing, including utility DAG failures (-fm-, -sm-, -tm-, -bdp-), Airflow platform instability or outage symptoms such as AirflowTaskTerminated or SIGTERM, and downstream Data Processing DAG failures caused by Airflow issues.

2) Scope

In Scope: Incidents created in ServiceNow due to Airflow alerts for c360-Prod and other environments if applicable; utility DAG failures and Airflow termination-related errors; routing and coordination with BDP.

Out of Scope: Incidents unrelated to Airflow or utility DAG patterns, which must be handled using the relevant SOP or runbook.

3) Systems & Links

ServiceNow (Incident Management): <https://itsmnow.service-now.com/>

Airflow UI: Environment-specific URLs (Prod / Non-Prod)

Jira Project: Project key / board link

BDP Support DL / Teams channel: Contact details

4) Ownership & Roles

BDP Team (Owner): Owns Airflow and Data Processing infrastructure; restores Airflow if down or unhealthy; takes corrective platform actions; confirms platform health post-recovery.

Support / Operations Team (Responder): Triage incidents; routes and escalates correctly; restarts impacted DAGs after Airflow stability; provides updates and closure documentation.

5) Triggers / When to Use This SOP

Use this SOP when a ServiceNow incident indicates Airflow workflow failures, utility DAG patterns (-fm-, -sm-, -tm-, -bdp-), or errors such as AirflowTaskTerminated or 'Task received SIGTERM signal'.

6) Definitions

Utility DAGs: Operational workflows identified by -fm- (First Mile), -sm- (Second Mile), -tm- (Third Mile), -bdp-.

SIGTERM / AirflowTaskTerminated: System termination due to infrastructure disruption, worker restart, pod eviction, scaling, or platform instability.

7) Procedure

Step 1: Identify the incident in ServiceNow and capture incident number, descriptions, environment, DAG and task details. Review long description.

Step 2: Determine if the incident is a utility DAG issue by checking identifiers (-fm-, -sm-, -tm-, -bdp-).

Step 3: Check for Airflow termination symptoms such as AirflowTaskTerminated or SIGTERM.

Step 4: Identify environment as Production or Non-Production.

Step 5A: Production – Route to APL_UDAP: Data Processing with appropriate work notes.

Step 5B: Non-Production – Create Jira ticket and notify BDP with details.

8) BDP Ownership Actions

Validate Airflow availability and infrastructure health. Restore platform if down or apply corrective actions if not down.

9) Post-Recovery Actions

Restart impacted Data Processing DAGs manually from Airflow UI if they do not recover automatically after platform stabilization.

10) Incident Communication & Updates

Update the incident after triage, routing, recovery confirmation, and retrigger actions using the recommended update template.

11) Resolution & Closure Criteria

Airflow stable; utility DAGs healthy; impacted DAGs retriggered and completed; work notes include root cause, actions taken, and timestamps.

12) Exception Handling

If utility patterns or SIGTERM are not present, treat as application or pipeline issue and route to the correct owner after validation.

13) Appendix: Quick Triage Checklist

Open incident; read description; identify utility DAG and SIGTERM; determine environment; route accordingly; retrigger DAGs; update and close incident.